

# CDMA Capacity Analysis: Processing Gain and SIR Trade-off

Deriving Maximum CDMA Users and Analyzing the Effect of SIR

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# Introduction

CDMA (Code Division Multiple Access) allows many users to share the same frequency band.

Users are separated using unique spreading codes.

Processing gain helps reduce the effect of interference.

SIR (Signal-to-Interference Ratio) determines communication quality.

Higher SIR generally improves call quality but reduces network capacity.

## Given Parameters

Processing Gain (G) = 180

Required SIR = 7

New Required SIR = 10

Simplified capacity equation:

$$N_{\max} = G / \text{SIR}$$

$N_{\max}$  = Maximum theoretical number of users

## Case 1: Required SIR = 7

Given:  $G = 180$ ,  $SIR = 7$

$$N_{\max} = G / SIR$$

$$N_{\max} = 180 / 7$$

$$N_{\max} = 25.714$$

Therefore, maximum whole-number users = 25

Theoretical capacity  $\approx 25$  users.

## Case 2: Required SIR = 10

Given:  $G = 180$ ,  $SIR = 10$

$$N_{\max} = G / SIR$$

$$N_{\max} = 180 / 10$$

$$N_{\max} = 18$$

Therefore, maximum users = 18.

Increasing the SIR requirement reduces capacity.

## Percentage Change in Maximum Users

Initial capacity:  $N1 = 180 / 7 = 25.714$

New capacity:  $N2 = 180 / 10 = 18$

% Change =  $((N2 - N1) / N1) \times 100$

% Change =  $((18 - 25.714) / 25.714) \times 100$

% Change =  $-30\%$

Capacity decreases by 30%.

# Comparison

Processing Gain: 180 in both cases

$SIR = 7 \rightarrow 25.714$  theoretical users  $\rightarrow 25$  whole users

$SIR = 10 \rightarrow 18$  theoretical users  $\rightarrow 18$  whole users

Capacity reduction = 30%

Key observation: Higher SIR  $\rightarrow$  Lower CDMA capacity

## Why Higher SIR Reduces Capacity

SIR compares desired signal strength with interference.

A higher SIR requirement means each user needs a cleaner signal.

More interference must be avoided or controlled.

Fewer users can share the same CDMA resources.

Therefore: Better signal quality  $\rightarrow$  Higher SIR  $\rightarrow$  Lower capacity.



# CDMA Capacity Trade-off

Lower SIR requirement:

- More users and higher capacity
- More interference may be tolerated
- Potentially lower call quality

Higher SIR requirement:

- Fewer users and lower capacity
- Better interference protection
- Better call quality

## Effect on Network Design

Operators continuously balance capacity and quality.

Power control can reduce unnecessary interference.

Cell and coverage planning can improve interference conditions.

Admission control prevents excessive loading.

Handoff and resource management help maintain service quality.

The objective is acceptable quality with maximum practical capacity.

The calculation is an idealized theoretical capacity estimate.

Practical capacity can be lower because of:

- Background noise
- Inter-cell interference
- Imperfect power control
- Multipath propagation
- User mobility and traffic activity
- Required engineering margins

# Conclusion

With  $G = 180$  and  $SIR = 7$ , theoretical capacity =  $180/7 = 25.714$  users.

With  $SIR = 10$ , theoretical capacity =  $180/10 = 18$  users.

Capacity decreases by 30%.

Higher  $SIR$  improves communication quality but reduces the number of users that can be supported.

This capacity-versus-quality balance is a defining CDMA design trade-off.

# References

Theodore S. Rappaport, Wireless Communications: Principles and Practice

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CDMA cellular communication and capacity concepts

Wireless network interference and capacity management concepts